$\therefore E_c = \mathbf{17.4\ V}$

Q5(a)

The energy of each photon of the electromagnetic radiation absorbed must be equal or larger than the work function of the metal which is the minimum energy needed to liberate an electron from the surface of the metal.

The excess energy becomes the KE of the liberated electron.

Q5(b)(i)

Maximum KE = eV_s

$= (1.6 \times 10^{-19})(2.3) = \mathbf{3.68 \times 10^{-19}\ J}$

Q5(b)(ii)

$hf = eV_s + \phi$

$$f = \frac{eV_s + \phi}{h}$$

$$= \frac{1.60\times10^{-19}\left(2.3+2.0\right)}{6.63\times10^{-34}}$$

= 1.04×10^{15} Hz

Q5(c) and (d)

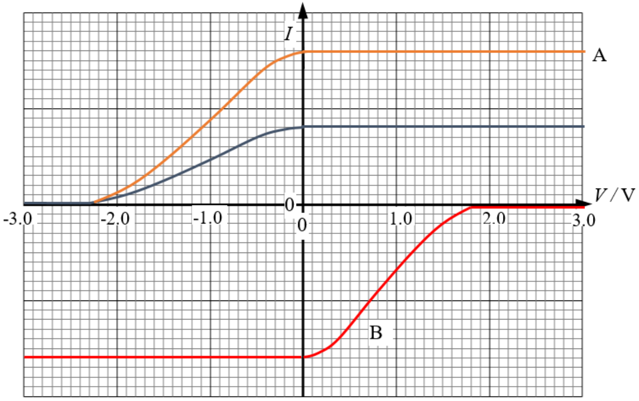

Fig. 5.2

Q6(a)(i)